

NRES 746 – Week 4, Lecture 2: Expectation and the Sample Mean

Name: _____

Instructions: Work alone for about 8-10 minutes. Then compare with a neighbor.

Part 1: A parameter vs. a statistic

Suppose the length of brook trout in a stream follows some distribution with true mean μ and variance σ^2 – these describe the *entire population* of trout in the stream, whether or not anyone ever catches one.

You catch $n = 5$ trout, measure them, and compute their average length $\bar{x}$.

1a. If a second team of researchers catches a *different* sample of 5 trout from the same stream and computes their own average, would you expect their $\bar{x}$ to come out exactly the same as yours? What about μ – does it change from team to team?

1b. Based on your answer to 1a: is μ a random variable, or a fixed number? Is $\bar{x}$ a random variable, or a fixed number? (A quantity is a random variable if it could come out differently on repetition.)

Part 2: Warming up with linearity of expectation

Two useful facts, true for any random variable X and constants a, b :

$$E[aX + b] = aE[X] + b \quad \text{Var}(aX) = a^2\text{Var}(X)$$

Suppose trout length X has $E[X] = 22$ cm and $\text{Var}(X) = 16$ cm².

2a. What is $E[2X]$? What is $\text{Var}(2X)$?

$E[2X] =$ _____ $\text{Var}(2X) =$ _____

2b. A second, independent measurement Y of a different trout has the same mean and variance as X . It is always true that $E[X + Y] = E[X] + E[Y]$, regardless of independence. Is $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ *always* true, or does it require something extra? What does it require?

Part 3: Averaging two measurements

Let X_1 and X_2 be two independent measurements of the same quantity, each with mean μ and variance σ^2 . Define their average $M = \frac{X_1 + X_2}{2}$.

3a. Rewrite M as $aX_1 + bX_2$ (i.e. what are a and b ?), then use the facts from Part 2 to derive $E[M]$ and $\text{Var}(M)$ in terms of μ and σ^2 . Show every step.

3b. Using $\mu = 22$ and $\sigma^2 = 16$ (from Part 2), what is $\text{Var}(M)$ as a number? Compare it to $\text{Var}(X_1) = 16$ alone – did averaging two measurements make your estimate more or less variable?

$\text{Var}(M) =$ _____

Part 4: Generalizing to n measurements

Now generalize: let $X_1, X_2, \dots, X_n$ be independent measurements, each with mean μ and variance σ^2 , and define the sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.

4a. Following the same logic as Part 3 (now with n terms instead of 2), derive $E[\bar{X}]$ and $\text{Var}(\bar{X})$ in terms of μ , σ^2 , and n .

4b. Check your formula against $n = 5$, $\sigma^2 = 16$: what number does $\text{Var}(\bar{X})$ come out to? Does it match Part 3b when you plug in $n = 2$?

$\text{Var}(\bar{X}) =$ _____

4c. As $n \rightarrow \infty$, what happens to $\text{Var}(\bar{X})$? What does that mean, in plain language, about how precisely you can pin down μ with a very large sample?

4d. One more thing to be careful about: $\text{Var}(\bar{X})$ and the *standard deviation* of $\bar{X}$ (often called the standard error) are not the same formula. If $\text{Var}(\bar{X}) = \sigma^2/n$, write down the standard deviation of $\bar{X}$ – how does it scale with n ?
